How to calculate a player’s “gravity”?

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What is a player's gravity?

- NBA.com claims that gravity is **"how much defensive pressure an offensive player draws both on and off the ball"**
  - Freeing up space for teammates
  - Give teammates open shots
  - Opening passing and driving lanes
  - Shifting defensive rotations

https://www.nba.com/news/intro-to-gravity-stat-nba-2025

- NBA measures gravity by comparing the expected **defensive pressure** and actual defensive pressure based on position of player on the court
  - $G = DPS_{actual} - DPS_{expected}$

Hypothesis and Propositions

A player's offensive gravity is associated with measurable changes in their teammates' **shot quality** and **offensive efficiency** when that player is on the court compared with when they are off.

1. Shot opportunities: Teammates attempt a higher proportion of open shots when a player who attracts significant defensive attention is on the court.
2. Offensive performance: Teammates have higher effective field goal percentages during the player's on-court minutes than during their off-court minutes.

P1: Shot Opportunities

Open Shot Percentage

Open%= ([OpenFGA + WideOpenFGA]/FGA)*100

OpenFGA: Teammates' field goal attempts with the nearest defender 4–6 feet away

WideOpenFGA: Teammates' field goal attempts with the nearest defender at least 6 feet away

FGA: Total field goal attempts by teammates

On/Off court Difference

$\Delta Open\% = Open\%_{on} - Open\%_{off}$ Positive difference supports P1.

P2: Offensive Performance

Effective Field Goal Percentage

eFG% = ([FGM+0.5*3PM]/FGA) *100

FGM: Field goals made

3PM: 3 points made

FGA: Field goal attempts

On/Off difference

$\Delta eFG\% = eFG\%_{on} - eFG\%_{off}$

A positive difference would support proposition 2.

Clutch Performance

Clutch game if less than 5% differential in last 5 minutes

Look at gravity and % of teams points is in clutch time

Scoring points in final momentis worth 10 points towards gravity (smtn like that)

How many times in a game does op[onetnt team tak ea time out

Assists to gravity

Create our own gravity metrics

Winshare of a player (team wins in a season / all players that played that season) (how important is a particular player to a team)

Random forest is fine. Maybe use multiple different models and compare?

Compare gravity to NBA: https://www.nba.com/inside-the-game/player/gravity?Season=2025-26&SeasonType=Regular+Season

On/Off Court Comparison

Offensive Rating

$ORtg = 100(PTS_s \div POSS_s)$

PTS_s: Team points scored during status s

$POSS_s$: Team offensive possessions during status s

Difference in Offensive Rating

$\Delta ORtg = ORtg_{on} - ORtg_{off}$

$ORtg_{on}$: Offensive rating while the player is on the court

$ORtg_{off}$: Offensive rating while the player is off the court

Further Considerations

How many of a given player's baskets are followed by an opposing team's time-outs?

Changing the shot charts of players he defends.

Plan

1. Pick players and get their on and off data.
2. Compare data.

Evaluate ATO plays in NBA

- Types of plays after time out:
  - Pick-and-rolls
  - Isolations
  - Off-ball screens
  - Handoffs
  - Post-ups
  - Cuts
- We are not sure how to find the videos yet